

Karakter: 12!!!

Energy Consumption

Eray Erkul 2026

Introduction for the Project

- RAPL vs. External measurements
 - How did we collect data?

Overall Question:

How accurate and precise are RAPL-based energy measurements compared to external power measurements, and does their performance depend on software characteristics?

Precise Research Question:

Do RAPL-based energy measurements differ systematically from external power measurements for CPU-bound and memory-bound workloads?

Controlled Variables & Experimental Setup

Device: Same hardware: Dell Latitude Laptop (Intel Core i5-8350U, 16GB RAM, NVMe Storage 512 GB).

OS: Same OS: Ubuntu Server 24.04 LTS.

Stress test Tool: **stress-ng** used to stress test (CPU vs. Memory).

Duration: 20-seconds but data was normalized to a fixed 20-second for a fair comparison. Assignment 2 had a 10 second pause.

Background services: disabled (excluding dbus service)

Cooling and environment: Same surface temperature

RAPL

Power supply: Powered by an outlet.

Sample size: 400 for each group
(Determined via course (Cochran's formula)).

Independent variable:

Stress test:

- Stress-ng - class: CPU & Memory

Dependent variable:

Energy consumption

External measurements

Power supply: Powered by a programmable DC power supply (SPD33303X-E by Siglent).

Sample size: 35 for each group
(Determined via time constraint).

A controlled setup was not constructed!

Analysis: Assignment 1

Issues with the obtained data:

- Different duration values
- Negative values
- Outliers

Features	Plots (CPU & Memory)	Shapiro-Wilk (CPU & Memory)	D'Agostino K ² (CPU & Memory)
CPU	Not normal distribution	0 out of 4	0 out of 4
PKG	Appears not to be a normal distribution	0 out of 4	0 out of 4

Permutation Test:

- Rejects the null hypothesis

Alternative Hypothesis (H_1):

There is a **significant difference** in energy consumption. One of the group (CPU or Memory) consumes more energy.

From plots and data -> CPU consume more energy.

Analysis: Assignment 2

Issues with the obtained data:

- Different duration values
- Outliers (Aggregation)

Features	Plots (CPU & Memory)	Shapiro-Wilk (CPU & Memory)	D'Agostino's K ² (CPU & Memory)
Energy	Does not look normally distributed	1 out of 2	2 out of 2
Power_avg	Looks somewhat normally distributed	2 out of 2	2 out of 2
Payload (Power)	Does not look normally distributed	2 out of 2	2 out of 2

Feature	Levene's Test (stat, p)	t-test (t, p)	ANOVA (stat, p)
Energy	Cannot reject null: all groups probably have equal variances (stat=0.241, p≈0.625)	Cannot reject null: means are probably equal (t=-0.318, p≈0.752)	Probably the same distribution (stat=0.101, p≈0.752)
Power_avg	Reject null: at least one pair of groups has unequal variances (stat=12.207, p≈0.00087)	Reject null: means are probably different (t=36.036, p≈0)	Probably different distributions (stat=1298.601, p≈0)
Payload	Cannot reject null: all groups probably have equal variances (stat=0.211, p=0.647)	Cannot reject null: means are probably equal (t=-0.493, p=0.624)	Probably the same distribution (stat=0.243, p=0.624)

Energy: There is **no statistically significant difference** in energy consumption between DRAM-heavy and CPU-bound workloads.

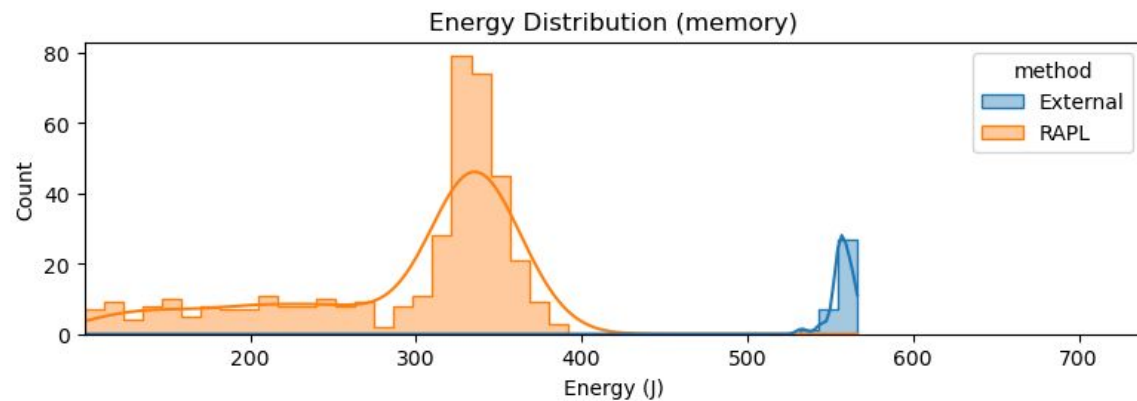
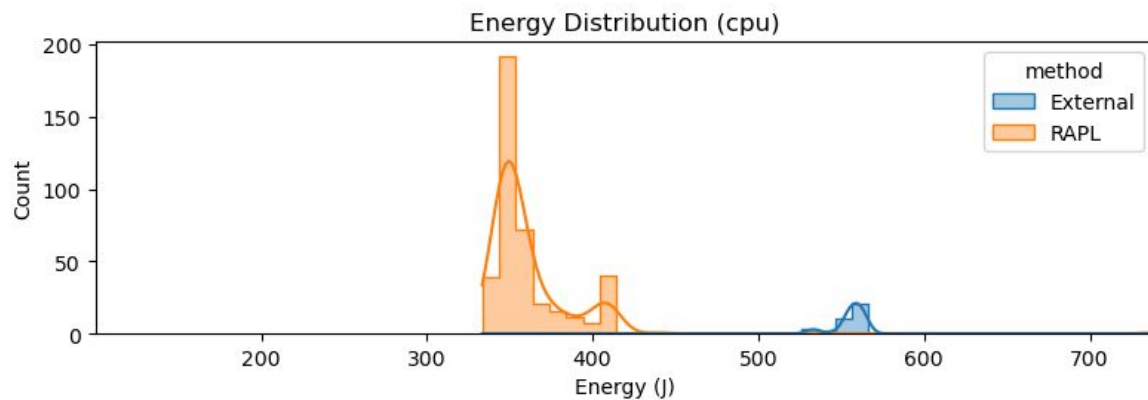
Power_avg: There is a **statistically significant difference** in average power consumption between DRAM-heavy and CPU-bound workloads.

Payload: There is **no statistically significant difference** in payload between DRAM-heavy and CPU-bound workloads.

Analysis: Mini project

	Energy Consumption	Accuracy	Precision
Hypothesis	H₀ (Null hypothesis): There is no significant difference in normalized energy consumption between CPU-bound and memory-bound workloads. H₁ (Alternative hypothesis): There is a significant difference in normalized energy consumption between CPU-bound and memory-bound workloads.	H₀ (Null hypothesis): RAPL measurements are as accurate as External measurements for total system energy consumption. H₁ (Alternative hypothesis): RAPL measurements are less accurate than External measurements for total system energy consumption.	H₀ (Null hypothesis): RAPL measurements are as precise as External measurements meaning the variance of normalized energy measurements do not differ between methods. H₁ (Alternative hypothesis): RAPL measurements are less precise than External measurements.
Statistical Results	External measurements: - Mann–Whitney U $p \approx 0.7246$ RAPL measurements: - Mann–Whitney U $p \approx 0.0000$	CPU: - Mann–Whitney U $p \approx 0.0000$ - RAPL underestimating energy usage by approximately 195.1 J. Capturing $\approx 65\%$ of External Memory: - Mann–Whitney U $p \approx 0.0000$ - RAPL reports around 264.7 J less energy, capturing only $\approx 52\%$ of External.	CPU: Levene's test $p \approx 0.0349$ RAPL variance: 790.06 External variance: 70.69 Difference (RAPL – External): 719.38 Memory: Levene's test $p \approx 0.0000$ RAPL variance: 5326.55 External variance: 43.19 Difference (RAPL – External): 5283.36
Decision	External (system level): Fail to reject H₀ RAPL (component level): Reject H₀	Reject H₀ for both CPU and memory RAPL is not accurate for measuring total system energy consumption.	Reject H₀ for both CPU and memory RAPL is less precise/consistent than External measurements.

Energy Distributions per Workload Class

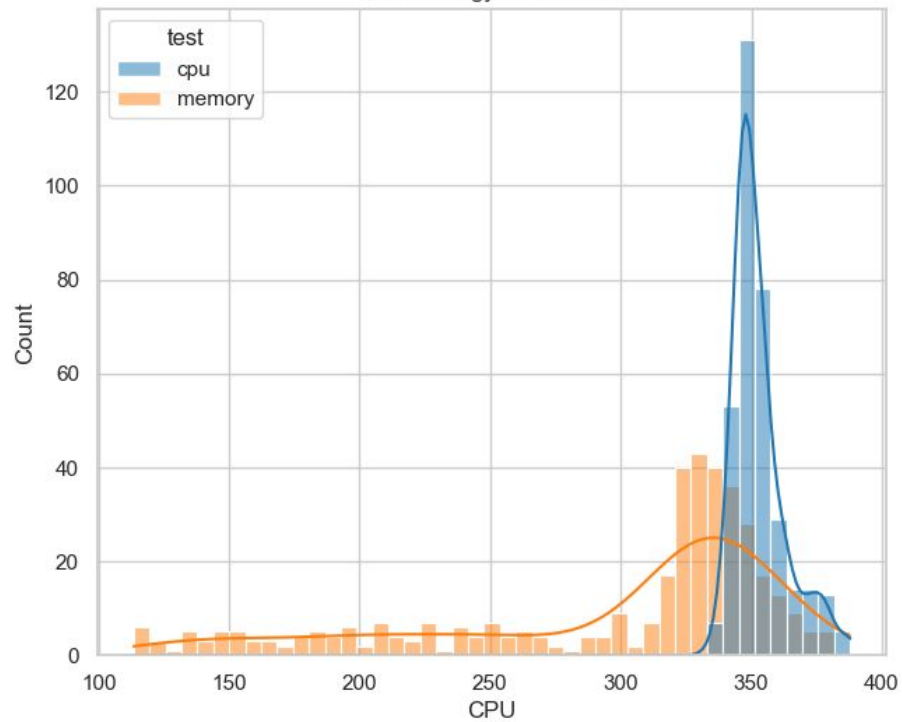


Validity

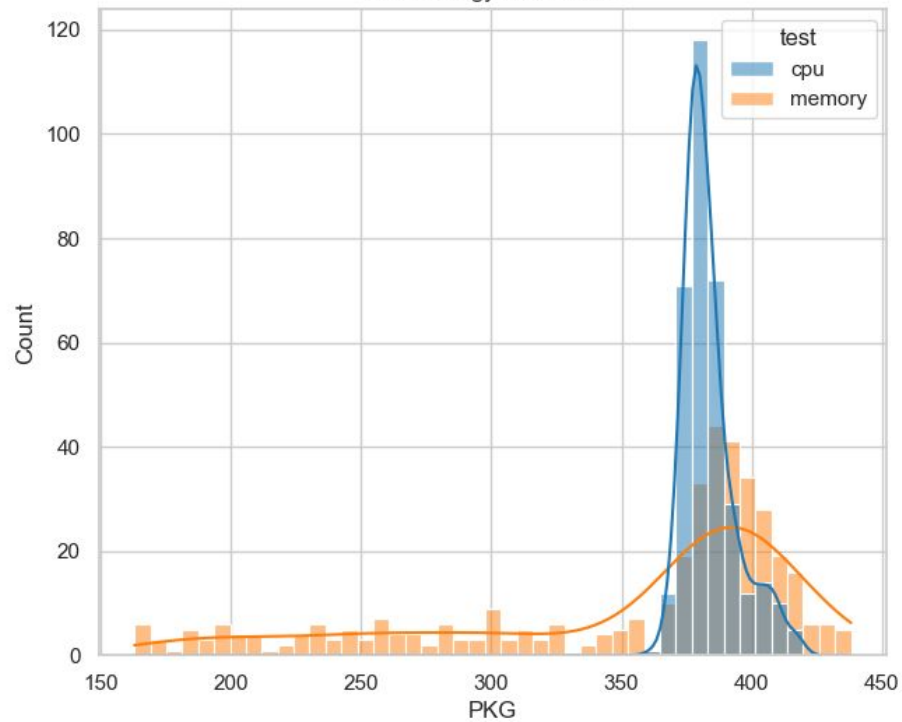
Confounding variables	Internal validity	External validity
• Negative values ◦ Measurement or calculation problems • Temperature issue • Maybe background services are not fully disabled • Wrong measured data has been appended into the csv file.	• Background processes running during tests • Execution time error (not precisely 20 seconds) • Measurement noise	• Hardware ◦ Old vs New hardware ◦ Energy efficiency • Experiments on other devices

Where else can RAPL be used? In business or school projects?

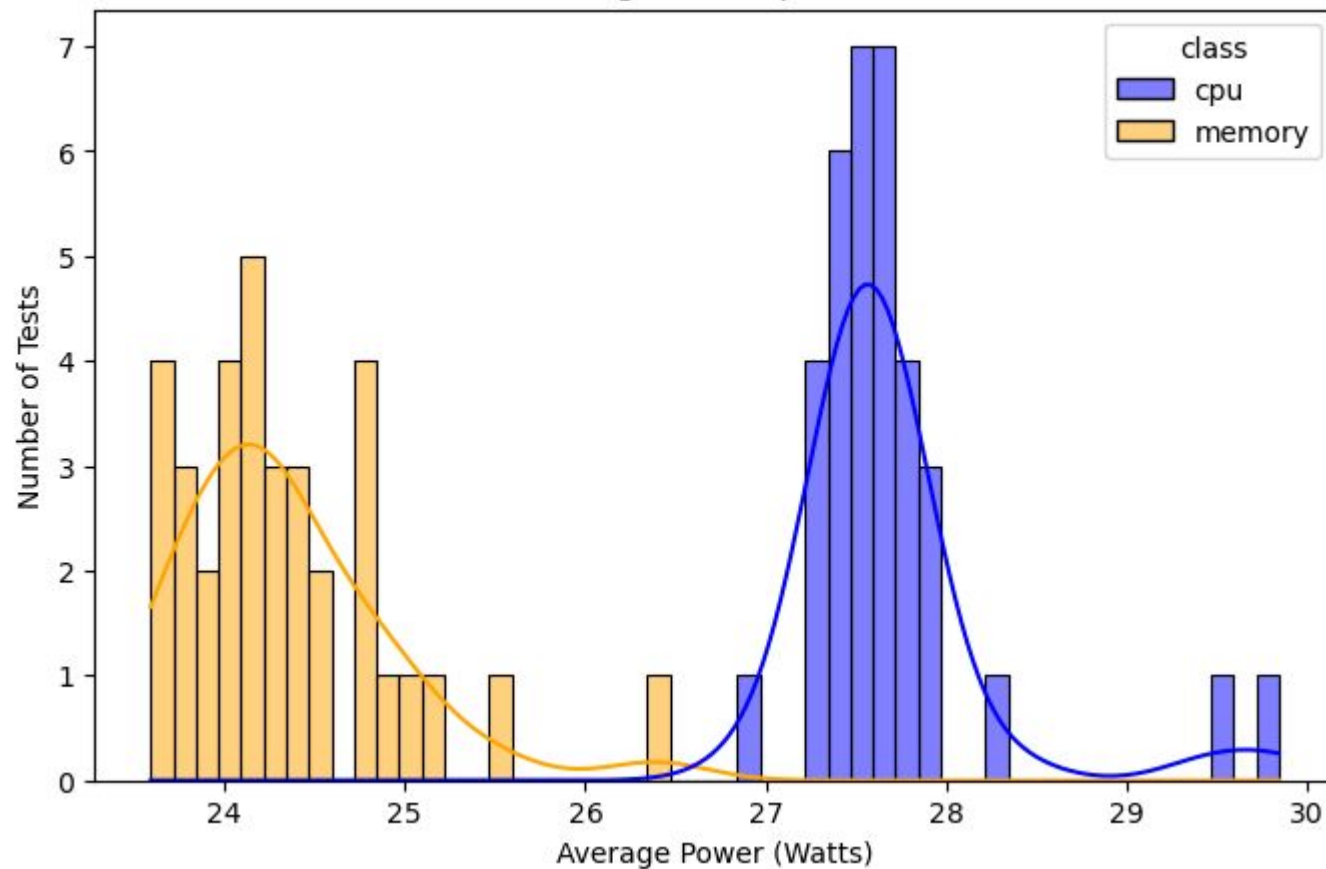
CPU Energy Distribution

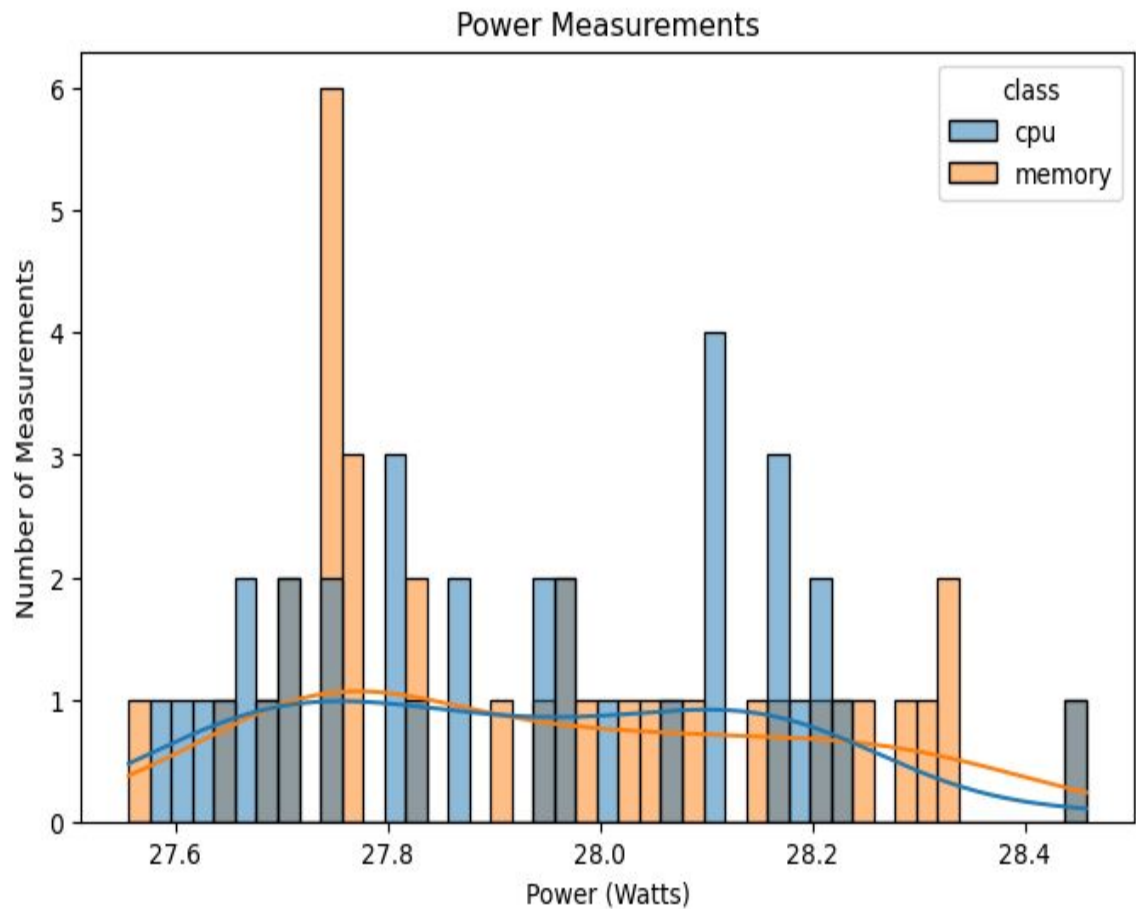
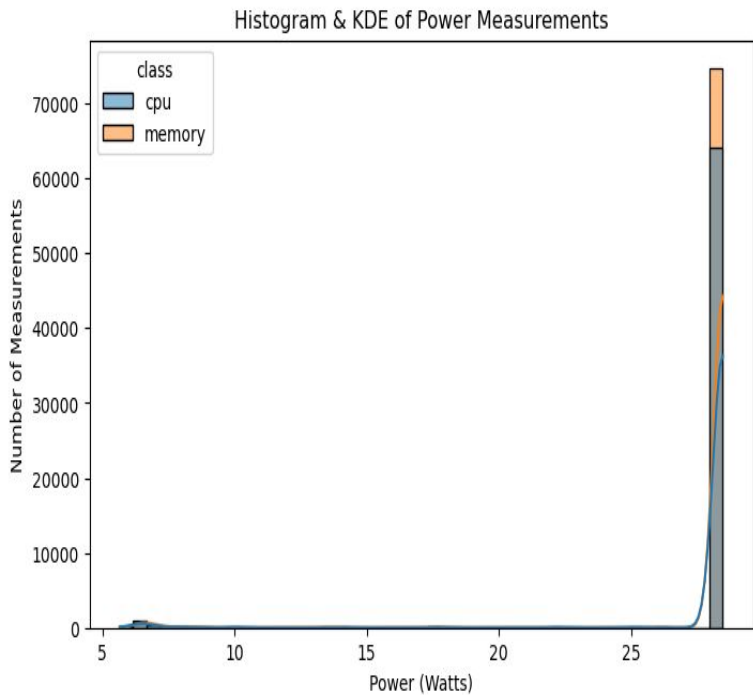


PKG Energy Distribution

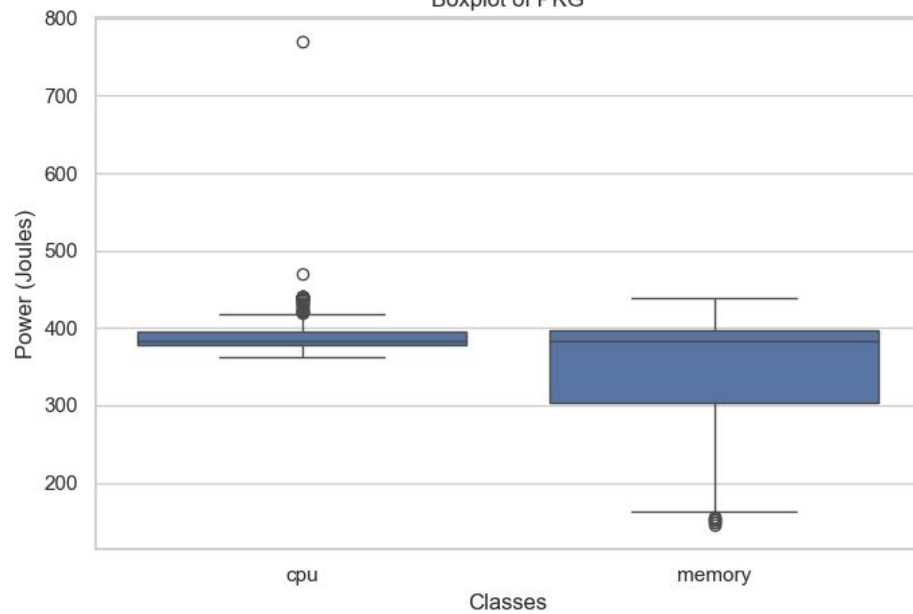


Average Power per Test

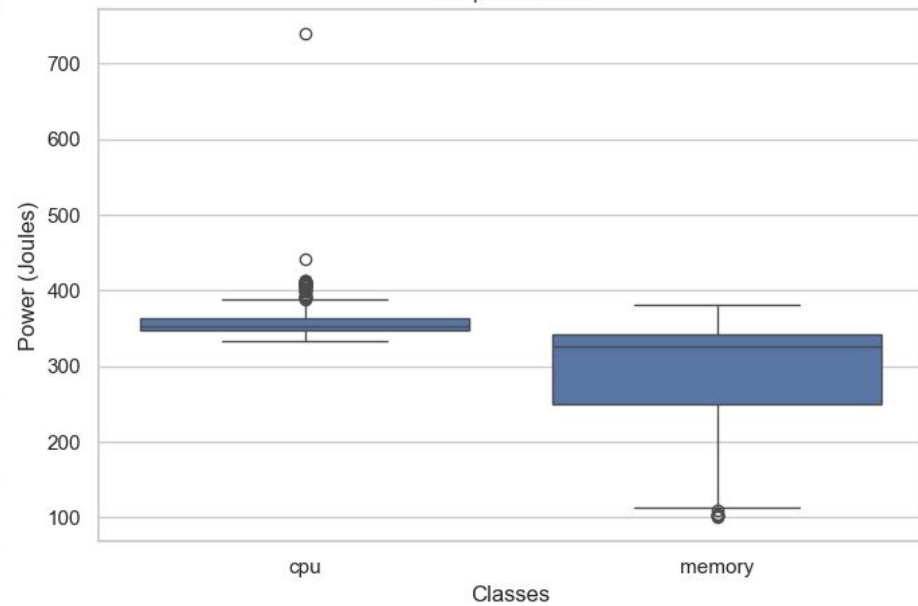




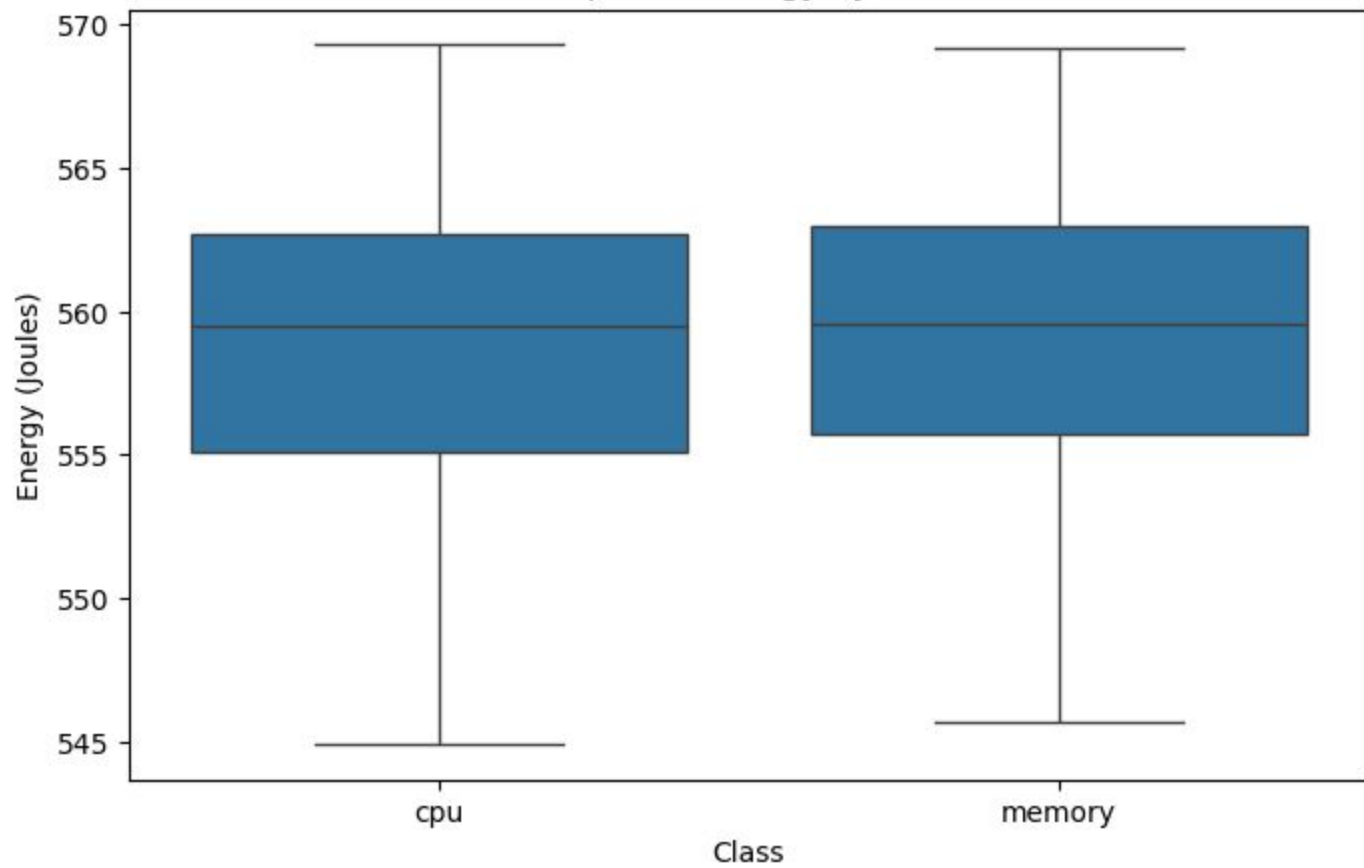
Boxplot of PKG



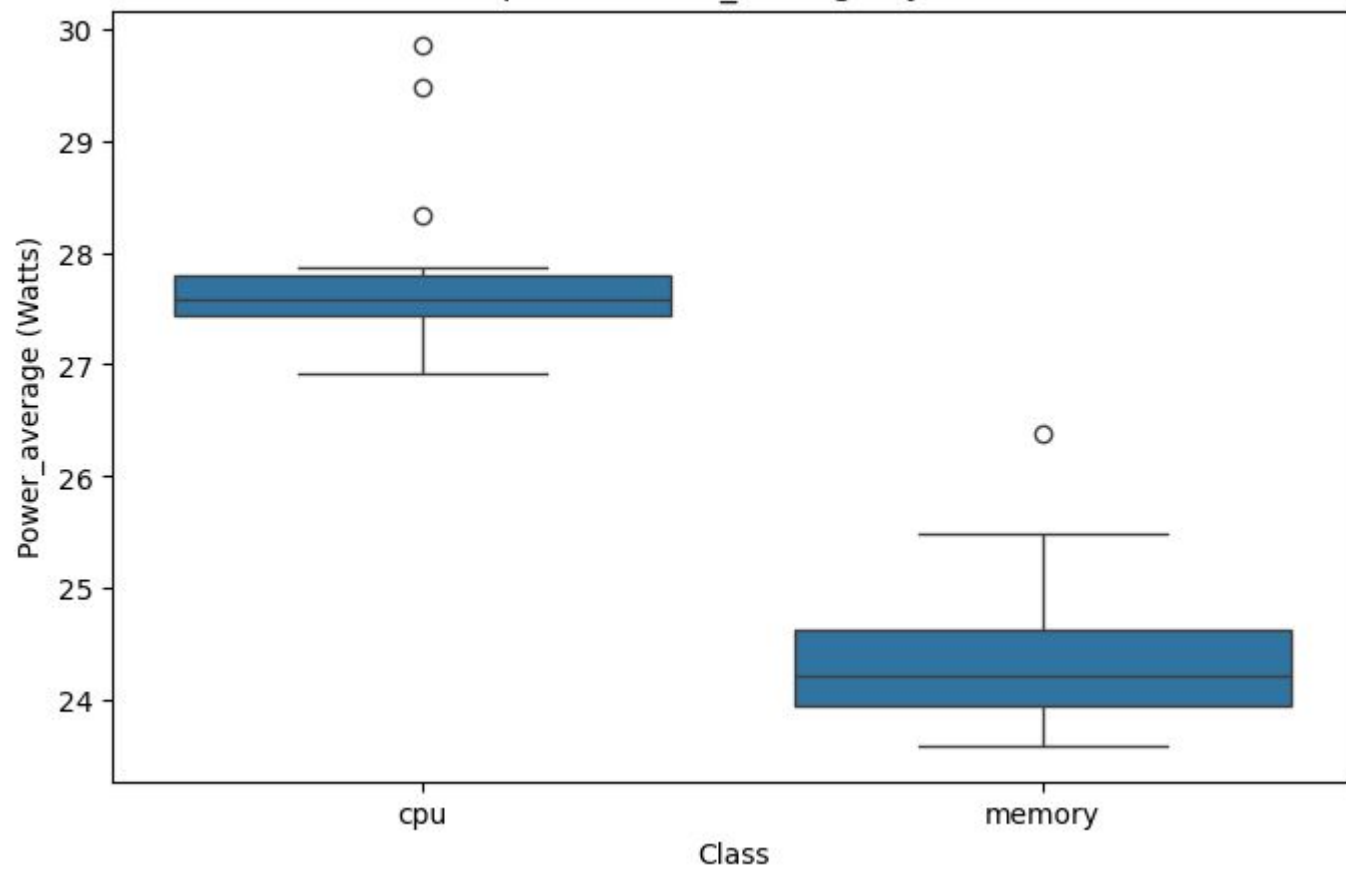
Boxplot of CPU



Boxplot of Energy by Class



Boxplot of Power_average by Class



Payload

